Statistical foundations of business analytics

Day 3 Cohorts and time-to-event analysis

Retention, censoring and Kaplan-Meier analysis

28 October 2026
Lecture slides for a 150 minute session

The lecture route

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Which acquisition channel retains customers better

Channel	New subscribers	Observed first cancellations
A	100	10
B	100	20

Is Channel A better simply because it has fewer cancellations?

What information about time is missing?

Follow-up time changes what those counts mean

Channel A launched last week. Channel B launched six months ago.

The second group has had much longer to experience and record a cancellation.

A defensible comparison specifies the same event, a common elapsed age and adequate observation in both groups.

ShopEase is fictional. All customer data and business results in this deck are synthetic teaching examples.

Business questions and suitable metrics

Question	Starting point
Who returns during later months	Period activity cohort table
How long until first cancellation	Time to first cancellation
How quickly leads first buy	Time to first qualifying purchase
Does onboarding improve retention	Randomized intervention with a common horizon

Day 3 adds elapsed time and observation length to the population, outcome and denominator defined on earlier days.

Forty purchases among one hundred leads

A report says 40 of 100 eligible leads purchased within seven days.

Thirty of the leads entered only yesterday.

Can we report a completed seven-day conversion rate of 40% for all 100?

Recent entrants lack a completed seven-day window

The arithmetic 40 / 100 equals 40%. The intended seven-day metric also requires seven observable days for each included lead.

Use an explicitly mature cohort for a simple completed-window rate. A survival analysis can instead use each lead's actual follow-up under its assumptions.

A funnel identifies a transition. Time-to-event analysis also describes how long the transition takes.

A cohort shares a defined entry condition

Here, a cohort contains customers whose first membership activation occurred in the same calendar month.

The entry rule is activation. Later product use is the activity we measure.

A customer stays in the original activation cohort when they return in a later month.

The January to April activity table and the May cancellation dataset are separate examples.

The activity retention metric contract

Component	Definition
Population	First-time activations, excluding test accounts
Unit	One customer, once per activity month
Qualifying activity	At least one product-use event in the month
Denominator	Original activation cohort size
Month 0	Activation month, with everyone qualifying at entry
Data cutoff	End of 30 April

One customer generates twenty use events

A January member uses ShopEase twenty times during February.

How much does this person contribute to the February active-customer numerator?

The person contributes one active customer

$$R_{c,k} = \frac{\text{customers from cohort } c \text{ active in period } k}{\text{customers originally in cohort } c}$$

Twenty use events still represent one active customer in that month.

Counting all twenty events would describe activity volume. The retention numerator counts distinct customers.

Calendar-month offsets and elapsed-day windows

Activation	Month 1 begins	Elapsed age on 1 February
1 January	1 February	31 days
31 January	1 February	1 day

Month 1 means the calendar month after activation. It is convenient for reporting but does not give everyone an identical 30-day interval.

A precise day-30 claim needs a window measured from each customer's activation.

Activity counts observed through the end of April

Cohort	Original	Month 0	Month 1	Month 2	Month 3
January	100	100	80	65	70
February	120	120	90	72	Not yet observable
March	80	80	64	Not yet observable	Not yet observable
April	100	100	Not yet observable	Not yet observable	Not yet observable

Columns show age periods relative to each activation cohort, rather than the same calendar month.

Reading across the January cohort

Calendar month	Cohort age	Active customers	Retention
January	Month 0	100	100%
February	Month 1	80	80%
March	Month 2	65	65%
April	Month 3	70	70%

Every percentage uses the original January denominator of 100.

Can January retention rise from 65% to 70%

January has 65 active customers in March and 70 in April.

Is an increase possible under the period activity definition?

Does it imply that nobody cancelled?

Customers can return after an inactive month

Yes. A member who was inactive in March can qualify as active in April.

Period activity measures use during the particular month. It does not require activity in every earlier month.

Activity and first cancellation are different outcomes. Reactivation cannot erase a first cancellation from a customer's history.

Activity retention by cohort age

Cohort size	Month 0	Month 1	Month 2	Month 3
January 100	100%	80%	65%	70%
February 120	100%	75%	60%	Not yet observable
March 80	100%	80%	Not yet observable	Not yet observable
April 100	100%	Not yet observable	Not yet observable	Not yet observable

Reading down Month 1 compares the same calendar-month offset. Signup timing, season and customer mix can still differ.

What belongs in April's Month 1 cell

The dataset closes at the end of 30 April.

Should April's Month 1 retention be 0%, 100% or not yet observable?

April's Month 1 is not yet observable

April activations reach Month 1 during May. That period lies beyond this report's cutoff.

A zero would assert that nobody returned during an observed period.

The report must distinguish no activity from no completed observation.

February Month 2 has seventy-two active members

The original February cohort contains 120 customers.

Calculate its Month 2 retention. Which calendar month does the numerator describe?

February Month 2 retention is 60%

$$R_{\text{February},2} = \frac{72}{120} = 60\%$$

The numerator is 72 distinct February activations who were active during April.

The denominator remains all 120 original February customers.

Pooled Month 1 retention across eligible customers

Cohort	Original size	Month 1 active
January	100	80
February	120	90
March	80	64
April	100	Not yet observable

Which cohorts belong in the denominator, and what is the pooled rate?

The pooled Month 1 rate is 78%

$$\frac{80 + 90 + 64}{100 + 120 + 80} = \frac{234}{300} = 78\%$$

January, February and March all have an observable Month 1. April does not.

Interpretation: 78% of eligible customers were active during their cohort's Month 1.

A customer-weighted rate and an equal-cohort average

$$\text{Equal-cohort average} = \frac{80 + 75 + 80}{3} = 78.33\%$$

The equal-cohort average gives each cohort one vote. The pooled rate gives each customer one vote.

For the fraction of eligible customers who were active, use 234 / 300 = 78%. Label an equal-cohort average explicitly.

Using 234 / 400 = 58.5% incorrectly adds immature April customers to the denominator.

Four different retention and conversion measures

Measure	What it counts
Period activity retention	Active in period k / original cohort
Consecutive activity retention	Active in every required period through k / original cohort
First-cancellation survival	Probability first cancellation occurs later than t
Fixed-window conversion	First purchase by a common horizon / eligible entrants

Changing the denominator to last month's active users creates another conditional transition metric.

The first-cancellation analysis contract

Choice	Definition
Population and unit	Eligible first membership episode per customer
Time origin	First paid membership activation
Event	First cancellation under one operational rule
Time unit	Days since the customer's own activation
Observation end	Event, loss of reliable tracking or analysis cutoff
Output	Probability of no first cancellation through day t

A cancellation request and an access end date differ

A member requests cancellation on day 10. Paid membership access ends on day 30.

Which date should define the event?

The business endpoint determines the timestamp

A question about cancellation requests uses the request date. A question about paid access ending uses the effective end date.

Use the chosen definition consistently for every customer and comparison group.

The main example uses first cancellation after activation under a fixed effective-cancellation rule.

At risk means still eligible for the first event

Just before a time, a customer is at risk if they are still observed and have not yet experienced the defined first event.

A high prediction score is unnecessary for membership in this risk set.

A previous first cancellation removes the person from later risk sets for that first-event analysis.

Everyone in the core examples is observed from their own activation.

One calendar cutoff with different elapsed durations

Customer	Activation	Observed end	Duration	End reason
A	1 May	3 May	2 days	Cancellation
B	7 May	10 May	3 days	Cutoff
C	2 May	6 May	4 days	Cancellation
D	3 May	7 May	4 days	Cancellation
E	5 May	10 May	5 days	Cutoff
F	1 May	7 May	6 days	Cancellation
G	2 May	10 May	8 days	Cutoff
H	2 May	10 May	8 days	Cutoff

The common analysis cutoff is 10 May. Each customer's elapsed clock starts at activation.

B has no cancellation during three observed days

B activates on 7 May. Reliable observation ends at the 10 May cutoff. No cancellation appears.

Is B a three-day cancellation, a permanent survivor or a right-censored record?

B is right censored after three days

We know that B remained free of first cancellation through three observed days.

We do not know B's eventual cancellation time or later event status.

Derived field	Value for B
Observed duration	3 days
Event indicator	0, because no event was observed

Observed duration and event coding

$$Y = \min(T, C), \qquad \delta = \mathbf{1}\{T \leq C\}$$

T is time to the first event. C is time to the end of reliable observation.

When the event is observed, record its elapsed time and status 1. Otherwise record the observation limit and status 0.

For a censored record, the true event time T remains unknown.

Three rows with the same coding rule

Customer	What happened	Duration	Status
A	Activated 1 May, cancelled 3 May	2	1
B	Activated 7 May, no event by 10 May	3	0
C	Activated 2 May, cancelled 6 May	4	1

An event after the frozen cutoff does not enter this version of the analysis. The person remains censored at the cutoff.

Data checks before a survival calculation

Keep one eligible first episode per customer and retain the original timestamps.

Check that activation precedes observation end and that every recorded event lies within follow-up.

Investigate zero durations. Immediate events, timestamp rounding and data errors need different handling.

An event known only to lie between inspections is interval censored. An inspection date adds an assumption about event timing.

A high-engagement group defined using the first month

An analyst labels customers high engagement if they use the product on at least ten days in the first month.

The analyst compares cancellation survival from activation against everyone else.

Why does this group definition create a problem?

The group label uses future information

At activation, first-month engagement is unknown. Early cancellation can also prevent a customer from qualifying.

Use groups defined by entry-time attributes, or randomize a specified onboarding intervention at entry.

A prespecified landmark analysis starts later among those still eligible. It answers a conditional question and needs an explicit label.

The censoring assumption behind Kaplan-Meier

Those who stop being observed should not have systematically different future event prospects from comparable people who remain observed.

Record why observation ended. A planned cutoff and a loss of tracking can have different implications.

A common cutoff can still coincide with cohort differences. Newer entrants have shorter follow-up and may behave differently.

A dissatisfied customer disables tracking

The customer's final recorded day contains no cancellation. Tracking then stops because the customer is dissatisfied.

Can we treat this loss as automatically equivalent to a planned reporting cutoff?

The reason for tracking loss may predict the event

Dissatisfaction may also predict future cancellation. The censored customers can differ from those still observed.

Audit loss reasons and tracking coverage. Examine whether the result changes under plausible assumptions about the missing follow-up.

Correct arithmetic and confidence intervals cannot repair selective observation.

Survival means remaining free of the defined first event

$$S(t) = P(T > t)$$

For first cancellation, S(30) = 0.80 means an estimated 80% have not first cancelled through day 30.

$$F(t) = P(T \leq t) = 1 - S(t)$$

The complement describes a first event by the stated horizon.

First-purchase survival at day 14 is 60%

The event is a first qualifying purchase, measured from lead creation.

$$S(14) = 0.60$$

What does the 60% represent? What proportion has purchased by day 14?

Sixty percent remain without a first purchase

$$P(T \leq 14) = 1 - 0.60 = 0.40 = 40\%$$

Survival means freedom from the event. Here, that event is a purchase.

A higher first-purchase survival curve means leads are remaining unconverted for longer.

The same survival value has different business meanings

First event	S(14) = 60%	1 - S(14) = 40%
Cancellation	No first cancellation through day 14	First cancelled by day 14
Purchase	No first purchase through day 14	First purchased by day 14
Setup completion	Setup still incomplete through day 14	Setup completed by day 14

Always name the event before calling a higher curve desirable.

A first-event survival curve cannot rise

Once a person has experienced a first cancellation, that historical first event has happened permanently.

The person may reactivate later. Current membership is a different business state.

Monthly use can rise after inactivity. Freedom from a first event can only stay unchanged or fall.

Event risk during days 7 to 14

$$S(7) = 0.80, \qquad S(14) = 0.60$$

Among customers still free of the event through day 7, what fraction experiences it during (7, 14]?

The conditional interval event risk is 25%

$$q_{7,14} = 1 - \frac{S(14)}{S(7)} = 1 - \frac{0.60}{0.80} = 25\%$$

Imagine 100 original customers. Eighty reach day 7 without the event, and sixty reach day 14.

Twenty events among the eighty earlier survivors gives 20 / 80 = 25%.

Conditional and original-population interval risks

Question	Calculation	Result
Event in (7,14] among original entrants	0.80 - 0.60	20%
Event in (7,14] among day-7 survivors	(0.80 - 0.60) / 0.80	25%
Survive to 14 among day-7 survivors	0.60 / 0.80	75%

The interval is identical. The conditioning group changes the denominator.

Hazard is an instantaneous conditional rate

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t < T \leq t + \Delta t \mid T > t)}{\Delta t}$$

If time is measured in days, hazard has units per day. It describes the instantaneous event rate among people still event-free.

A rate is not bounded by one. A probability always lies between zero and one.

Rate times a sufficiently short interval approximates a conditional probability. Daily hazard times thirty is not an exact 30-day risk.

Two cancellations among twenty at-risk members

$$\text{Recorded-day conditional event fraction} = \frac{2}{20} = 10\%$$

This is a discrete conditional proportion for a stated day. It differs from a continuous instantaneous hazard.

A high early risk can suggest where to investigate onboarding. Renewal timing or timestamp conventions can also create a peak.

A descriptive timing pattern does not establish its cause.

Five minute break

Kaplan-Meier uses the same conditional-probability idea repeatedly.

At each observed event time, the denominator contains people still event-free and under observation just before that time.

Multiplying survival through successive event times

Ten customers are observed without censoring. Two cancel at the first event time. One of the remaining eight cancels at the next.

$$\frac{8}{10} \times \frac{7}{8} = \frac{7}{10} = 70\%$$

Reach the first point without an event, then remain event-free at the next point.

Censoring changes later risk sets while preserving the earlier observed experience.

The Kaplan-Meier estimator

$$\widehat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

Symbol	Meaning at event time ti
ni	Observed and event-free just before the time
di	Number of first events at that time
1 - di / ni	Conditional fraction remaining event-free

Multiply across event times at or before t. Between event times, the estimate stays flat.

The eight-subscriber cancellation dataset

Customer	Duration in days	Event status	Observed endpoint
A	2	1	First cancellation
B	3	0	No first cancellation through day 3
C	4	1	First cancellation
D	4	1	First cancellation
E	5	0	No first cancellation through day 5
F	6	1	First cancellation
G	8	0	No first cancellation through day 8
H	8	0	No first cancellation through day 8

Status 1 means an observed first cancellation. Status 0 means right censoring.

Three tempting summaries of the eight records

Summary	Value
Observed events / customers	4 / 8 = 50%
Average of observed cancellation times	(2 + 4 + 4 + 6) / 4 = 4 days
Average of all recorded durations	40 / 8 = 5 days

Does any value directly estimate mean membership lifetime or day-8 cancellation probability?

Each summary answers a narrower question

Four of eight is the fraction with an event observed before their individual observation ends.

Four days averages only the observed cancellations. It excludes censored event times.

Five days averages event times and observation limits together.

Unequal follow-up prevents interpreting these as the intended lifetime or common-horizon probability.

The first cancellation occurs at day 2

All eight customers have durations of at least two days. A first cancels at day 2.

Who is at risk just before day 2? Calculate the survival factor and survival immediately after the event.

Day 2 leaves survival at seven eighths

$$n_2 = 8, \qquad d_2 = 1, \qquad \widehat{S}(2) = 1 \times \frac{7}{8} = 0.875$$

All eight contribute to the first risk set. A then leaves because the event occurred.

B, C, D, E, F, G and H remain under observation and event-free after day 2.

B reaches the observation cutoff at elapsed day 3

Seven customers remain at risk just before day 3. B's observation ends without an event.

Should the survival estimate fall? Who remains for day 4?

Censoring changes the next risk set

$$n_3 = 7, \quad d_3 = 0, \quad c_3 = 1, \quad \widehat{S}(3) = 0.875 \times 1 = 0.875$$

B contributes known event-free time through day 3. No event was observed at this time.

The day-4 risk set contains C, D, E, F, G and H. Its size is six.

C and D both cancel at day 4

The six people at risk just before day 4 are C, D, E, F, G and H.

Two events occur at this same recorded time. Calculate the factor and new cumulative survival.

Two events among six give a four-sixths factor

$$1 - \frac{d_4}{n_4} = 1 - \frac{2}{6} = \frac{4}{6}$$

$$\widehat{S}(4) = \frac{7}{8} \times \frac{4}{6} = \frac{7}{12} = 0.5833$$

C and D leave together. E, F, G and H remain after the events.

Tied events share the same risk set. The denominator is six, rather than the original eight.

E is censored at day 5 and F cancels at day 6

Four remain after day 4: E, F, G and H. E's observation ends without an event at day 5.

How many remain at risk just before day 6, and what is survival after F cancels?

Three remain at risk before the day-6 event

F, G and H remain after E is censored. Day 5 creates no downward step.

$$\widehat{S}(6) = \frac{7}{12} \times \frac{2}{3} = \frac{7}{18} = 0.3889$$

The day-6 event removes one of three. The day-2 event removed one of eight.

One event can produce a larger proportional drop when the later risk set is smaller.

G and H are both censored at day 8

The final two people remain event-free through eight observed days, then reach the cutoff.

Should survival now drop to zero or extend indefinitely?

The final estimate remains seven eighteenths

$$\widehat{S}(8) = \frac{7}{18} \times 1 = 0.3889$$

No cancellation occurs at day 8. Both records end through censoring.

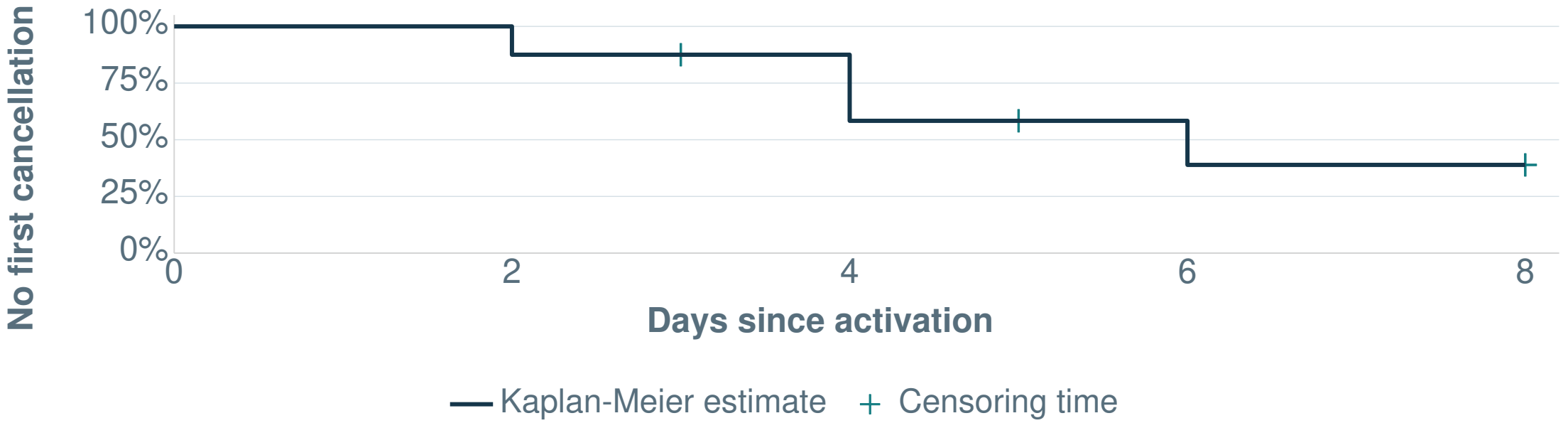
After day 8 nobody remains observed. The data do not support extending the curve beyond that point.

The complete event and censoring table

Day	At risk n	Events d	Censored c	Factor	Survival
0	8	0	0	1	1.0000
2	8	1	0	7 / 8	0.8750
3	7	0	1	1	0.8750
4	6	2	0	4 / 6	0.5833
5	4	0	1	1	0.5833
6	3	1	0	2 / 3	0.3889
8	2	0	2	1	0.3889

Risk sets are before events. Survival is after events. Later risk set = n - d - c.

Kaplan-Meier survival for the eight subscribers



Day	0	2	4	6	8
At risk just before day	8	8	6	3	2

Plus marks show censoring. The shared mark at day 8 represents G and H.

Reading the curve at a specified day

$$\widehat{S}(4) = 58.3\%, \qquad \widehat{S}(6) = 38.9\%$$

$$\widehat{F}(6) = 1 - \frac{7}{18} = \frac{11}{18} = 61.1\%$$

Under the censoring assumptions, the estimated probability of a first cancellation by day 6 is 61.1%.

The raw fraction of observed events is 50%. The common-horizon estimate accounts for different follow-up lengths.

What is the estimated median time to first cancellation

After event time	Estimated survival
Day 2	87.5%
Day 4	58.3%
Day 6	38.9%

At which first event time does the curve reach 50% or lower?

The survival median is six days

The curve stays above 50% at day 4 and first falls below it at day 6.

$$\widehat{t}_{0.5} = \inf\{t : \widehat{S}(t) \leq 0.5\} = 6 \text{ days}$$

The median of the four observed event times, 2, 4, 4 and 6, is four days. That calculation excludes censored experience.

A median that is not reached during follow-up

If survival remains above 50% throughout the supported follow-up, report that the median was not reached.

The final observed day is not a substitute median. A flat tail supplies no observed future event time.

Report a supported survival probability or restricted event-free time instead.

A member has already remained event-free for four days

$$\widehat{S}(4) = \frac{7}{12}, \qquad \widehat{S}(6) = \frac{7}{18}$$

Conditional on no first cancellation through day 4, what is the estimated chance of remaining free through day 6?

Conditional survival through day 6 is two thirds

$$\frac{\widehat{S}(6)}{\widehat{S}(4)} = \frac{7/18}{7/12} = \frac{2}{3} = 66.7\%$$

The corresponding conditional first-cancellation risk during (4, 6] is one third, or 33.3%.

The unconditional 38.9% starts at activation. This question starts with the group that reached day 4 event-free.

The same interval with two reference populations

$$\widehat{S}(4) - \widehat{S}(6) = \frac{7}{12} - \frac{7}{18} = \frac{7}{36} = 19.4\%$$

This is the estimated fraction of original entrants whose first cancellation occurs during (4, 6].

$$\frac{\widehat{S}(4) - \widehat{S}(6)}{\widehat{S}(4)} = \frac{1}{3} = 33.3\%$$

This is the fraction among those event-free through day 4.

The curve is flat from day 6 to day 8

Only G and H remain observed after the day-6 event. Neither has an observed cancellation before day 8.

Does the flat segment prove that population cancellation hazard is zero?

A flat estimate reflects the observed events

The estimate stays flat because no event was observed during this interval.

Only two people support this tail. Their experience cannot establish that the population event rate is zero.

The eight-person dataset teaches the calculation. Business decisions require a larger body of evidence and uncertainty estimates.

Evidence that makes a survival estimate credible

Issue	What the analyst examines
Small later risk sets	At-risk counts, confidence intervals and supported horizon
Selective tracking loss	Reasons, tracking audit and sensitivity analysis
Calendar or cohort change	Entry dates, promotions and comparable periods
Inconsistent event times	One event definition and a timestamp audit
Selected entry population	Eligibility and observation from the chosen origin

A survival result needs its analysis context

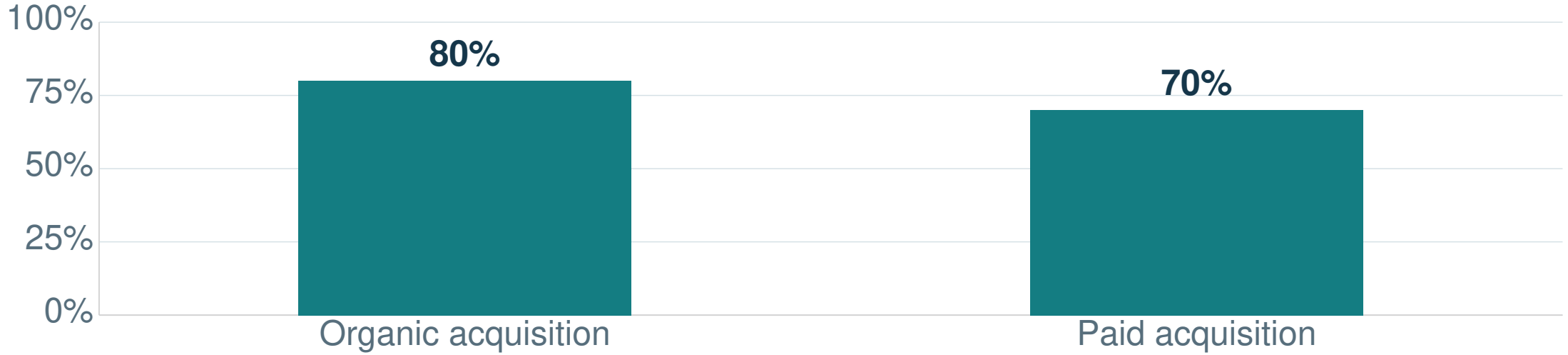
Report the population and entry dates, time origin, event rule, cutoff and time unit.

Give the customer count, event count, censoring pattern, horizon and numbers at risk.

State the uncertainty method and relevant assumptions.

A confidence interval quantifies statistical uncertainty under those assumptions. It does not correct misclassified events or selective observation.

Organic and paid acquisition at the same 30-day horizon



A separate, larger synthetic report estimates 80% and 70% freedom from first cancellation.

The survival difference is 10 percentage points. First-cancellation risks are 20% and 30%.

Would switching channel improve retention by ten points

Organic and paid groups have estimated day-30 survival of 80% and 70%. The groups were not randomized.

Does the difference establish what would happen if the business switched a customer's acquisition channel?

The comparison is descriptive

Plan, region, motivation, promotion and signup month can differ across the acquisition groups.

A common elapsed age addresses the time scale. It does not balance those other attributes.

Request confidence intervals and at-risk counts, inspect cohort and censoring differences, and test a specific actionable intervention.

A fixed-horizon effect and a log-rank test

Question	Useful result
How large is the difference at day 30	Survival difference at day 30 with uncertainty
Do survival experiences differ across follow-up	A suitable log-rank test
How is event rate related to measured attributes	A suitable survival regression model
Does an intervention improve retention	Randomized study with planned follow-up

The log-rank test combines observed versus expected events across event times. Its p-value is not a day-30 effect size.

A retention analysis should identify a measurable next step

Choose a business horizon that the follow-up supports in every comparison group.

Name the outcome and size of the difference, then describe the uncertainty and observation assumptions.

If the comparison is observational, use it to motivate a specific causal test such as an onboarding change.

Restricted mean survival time can summarize event-free days accumulated within a chosen horizon.

Restricted mean survival time

$$\mathrm{RMST}(\tau) = \int_0^{\tau} S(t)\,dt$$

This is the estimated event-free time accumulated within the first tau days.

For a step curve, add rectangle widths multiplied by their survival heights. No calculus is needed for the arithmetic.

It is a restricted summary within follow-up. A positive final tail does not identify unrestricted mean lifetime.

Adding event-free days up to day 8

Interval	Width	Survival height	Area in days
0 to 2	2	1	2.0000
2 to 4	2	7 / 8	1.7500
4 to 6	2	7 / 12	1.1667
6 to 8	2	7 / 18	0.7778
Total to day 8	8		5.6944

The estimate is 5.6944 event-free days within the first eight days. Group comparisons must use the same supported horizon.

One additional event-free day in a randomized trial

A separate onboarding trial estimates one additional event-free day per entrant within the first 30 days.

Each such day contributes 40 cents in margin. The intervention costs 20 cents per entrant.

What is the estimated incremental margin per entrant?

Estimated incremental margin is twenty cents per entrant

$$1 \text{ extra day} \times 40 \text{ cents/day} - 20 \text{ cents} = 20 \text{ cents}$$

The calculation assumes constant contribution per event-free day and no other business effects.

Uncertainty in the retention effect must also inform the decision. Variable margins, costs and reactivation can require a fuller outcome model.

Worked exercises

Exercise	Focus
A1	Cohort rates, maturity and pooling
A2 and A3	A first-purchase contract and future information
B1 to B3	A new risk table, curve and conditional probabilities
B4 and B5	Purchase timing and a business recommendation

The following worksheet datasets are separate from the earlier examples.

A1 Pooled activity retention with unequal cohort sizes

Cohort	Original size	Month 1 active
January	100	60
February	200	100
March	100	80
April	100	Not yet observable

Calculate the observed cohort percentages and pooled Month 1 rate. The cutoff is the end of April.

A1 The pooled customer rate is 60%

Cohort	Calculation	Month 1 retention
January	60 / 100	60%
February	100 / 200	50%
March	80 / 100	80%

$$\frac{60 + 100 + 80}{100 + 200 + 100} = \frac{240}{400} = 60\%$$

Four hundred customers belong to cohorts with observable Month 1 activity.

A1 Two alternative dashboard calculations

A dashboard reports 240 / 500 = 48%. Another analyst averages 60%, 50% and 80%.

Explain the 48% error. Calculate the equal-cohort average and compare it with the 60% pooled customer rate.

A1 The denominator and weighting explain the differences

The 500 denominator includes April's 100 customers even though their Month 1 is not yet observable.

$$\text{Equal-cohort average} = \frac{60 + 50 + 80}{3} = 63.33\%$$

The larger February cohort has lower retention and receives twice the customer weight. This brings the pooled rate to 60%.

A1 January activity rises in Month 2

The worksheet January cohort contains 100 original customers. Sixty are active in Month 1 and seventy in Month 2.

Can retention rise? Could first-cancellation survival rise in the same way?

A1 Period activity can rise from 60% to 70%

Yes. People can return after an inactive period and count in the Month 2 activity numerator.

A first-event curve cannot rise. A person who first cancelled remains someone with a first cancellation, even after reactivation.

The activity table alone does not establish cancellation status.

A2 Time from lead creation to first purchase

ShopEase freezes data on 30 April. Some leads entered in January and others on 29 April.

Define the population, unit, origin, event, event indicator and duration for a lead without a purchase.

A2 A valid first-purchase analysis contract

Field	Definition
Population	Eligible first-time leads with reliable tracking
Unit	One deduplicated lead or customer
Origin	Lead creation timestamp
Event	First qualifying completed purchase
Event status	1 if observed by cutoff, otherwise 0
No purchase by cutoff	Duration from creation to reliable observation end

A 29 April entrant has about one observed day, rather than a completed 30-day conversion window.

A3 First-month engagement as a baseline group

The high-engagement label requires at least ten use days during the first month. Survival is compared from activation.

What is unavailable at activation? What would make the starting comparison more defensible?

A3 Group definitions need the right time origin

The future engagement total is unknown at activation. Remaining active long enough can affect whether someone earns the label.

Use an entry-time attribute, or assign an onboarding intervention randomly at activation.

A properly defined later landmark compares people still eligible at that later time and answers a conditional question.

B1 A separate six-customer cancellation dataset

Customer	Duration in days	Event status
P	2	1
Q	3	0
R	4	1
S	5	0
U	7	1
V	9	0

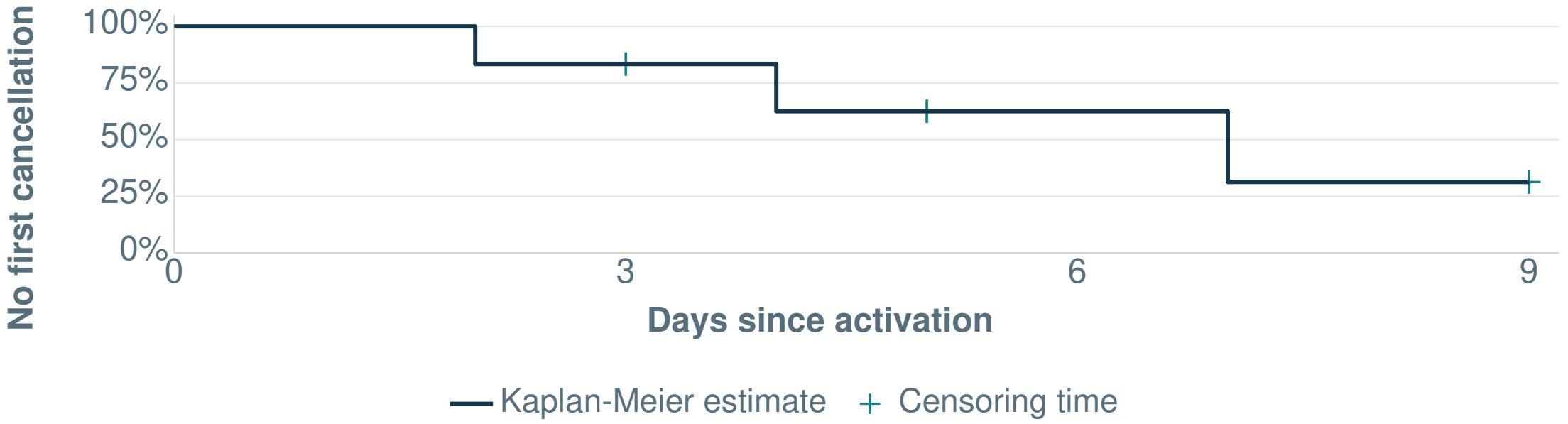
Build the risk table at days 2, 3, 4, 5, 7 and 9, using status 1 for events and status 0 for censoring.

B1 Risk sets and cumulative survival

Day	At risk n	Events d	Censored c	Factor	Survival
2	6	1	0	5 / 6	0.8333
3	5	0	1	1	0.8333
4	4	1	0	3 / 4	0.6250
5	3	0	1	1	0.6250
7	2	1	0	1 / 2	0.3125
9	1	0	1	1	0.3125

Each later risk set removes the previous events and censorings. Censor-only rows multiply survival by one.

B1 The six-customer step curve



Day	0	2	4	7	9
At risk just before day	6	6	4	2	1

Censoring occurs at days 3, 5 and 9. The data end at day 9.

B2 Survival, cancellation probability and median

Using the six-customer table, calculate survival through day 4 and day 7, and first-cancellation probability by day 7.

What is the survival median? Why is V's day-9 endpoint not a final downward step?

B2 Day-7 survival is 31.25%

$$\widehat{S}(4) = \frac{5}{6} \times \frac{3}{4} = \frac{5}{8} = 62.5\%$$

$$\widehat{S}(7) = \frac{5}{8} \times \frac{1}{2} = \frac{5}{16} = 31.25\%$$

First-cancellation probability by day 7 is 1 - 5 / 16 = 68.75%. The median is seven days, the first step to 50% or lower.

V is censored at day 9, so no event creates a step there.

B3 Conditional survival from day 4 to day 7

The six-customer dataset has S-hat(4) = 5 / 8 and S-hat(7) = 5 / 16.

Calculate conditional survival to day 7. Compare cumulative cancellation by day 7 with the raw observed event fraction 3 / 6.

B3 Conditional survival is 50%

$$\frac{\widehat{S}(7)}{\widehat{S}(4)} = \frac{5/16}{5/8} = \frac{1}{2} = 50\%$$

The common-horizon cumulative cancellation estimate is 68.75%. The raw event fraction is 3 / 6 = 50%.

Earlier censored customers contribute to early risk sets and leave later ones. Kaplan-Meier uses that observed experience under the censoring assumptions.

B4 The event changes to first purchase

$$\widehat{S}(7) = 0.60, \qquad \widehat{S}(14) = 0.40$$

What proportion has purchased by day 14? Among leads still unconverted through day 7, what proportion purchases during (7, 14]?

B4 Sixty percent purchased by day 14

$$1 - \widehat{S}(14) = 1 - 0.40 = 60\%$$

$$1 - \frac{\widehat{S}(14)}{\widehat{S}(7)} = 1 - \frac{0.40}{0.60} = \frac{1}{3} = 33.33\%$$

The unconditional interval conversion is 0.60 - 0.40 = 20% of original entrants.

The 33.33% denominator contains leads still unconverted at day 7.

B5 A recommendation from organic and paid acquisition

Estimated first-cancellation survival at day 30 is 80% for organic and 70% for paid acquisition. The groups were not randomized.

Give a descriptive conclusion, two evidence requests and one follow-up action.

B5 An evidence-based recommendation

Organic acquisition has ten percentage points higher estimated freedom from first cancellation at day 30 in these data.

Request confidence intervals and at-risk counts. Compare entry cohorts, plans, promotions and censoring patterns.

Test a specific onboarding improvement in a randomized study with a prespecified retention horizon.

Three review questions

1 What information does a right-censored record contribute?

2 Why can period activity retention rise while first-event survival cannot?

3 What must a day-30 retention comparison state?

The distinctions behind a defensible time-based metric

1 Known event-free time through the last reliable observation, with later event status unknown.

2 Customers can return to activity. A historical first event cannot be undone.

3 A common event definition, time origin and horizon, eligible populations, sufficient follow-up and uncertainty.

Individual event timing and calendar-time forecasting

Day 3 question	Day 4 question
How long until an individual first cancels	How many cancellations will occur next week
Clock starts at each person's activation	Clock follows calendar dates
Rows describe individual follow-up	Rows describe repeated calendar periods

The two views can inform each other, while their units and forecasting questions remain different.

Optional uncertainty formula

$$\widehat{\mathrm{Var}}\{\widehat{S}(t)\} = \widehat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{n_i(n_i - d_i)}$$

Greenwood's expression helps explain why small later risk sets affect precision.

Boundary cases need appropriate software handling. Probability limits commonly use a transformation rather than a naive symmetric interval.

The formula is optional and is not needed for the worked exercises.

Optional conventions for tied times and delayed entry

If events and censorings share a recorded time with unknown ordering, a common convention includes tied censored observations in the event risk set, then removes them.

Use finer timestamps when available and state the convention. The main example has tied events but no event-censoring tie.

Customers first observed partway through membership need delayed-entry risk sets if the clock remains activation. They cannot contribute to earlier unobserved risk sets.

Boundaries of this first-event analysis

Repeated cancellations require a clear episode definition or a recurrent-event or multistate method, with dependence handled appropriately.

For one termination cause when other terminations prevent it, use competing-risk cumulative incidence. Censoring other causes and taking 1 - S-hat generally does not estimate actual cause probability.

Extrapolating lifetime or lifetime value beyond observed follow-up requires additional assumptions and validation.

Optional interpretation of a hazard ratio

$$\text{Hazard ratio} = 0.8$$

Under an appropriate proportional-hazards model, this describes a 20% lower instantaneous event rate conditional on the included covariates.

It does not directly specify a 20-percentage-point retention benefit or a 20% longer lifetime.

An adjusted association is causal only with a defensible causal design and assumptions. Crossing curves can also weaken the usefulness of a single summary.

Optional spreadsheet setup

Columns	Contents	Rows
A, B, C	Customer ID, duration, event status	2 to 9, from the eight-customer example
E	Time	2, 3, 4, 5, 6, 8 in E2:E7
F, G, H	At risk, events, censored	One row per listed time
I, J	Survival factor, cumulative survival	Formulas filled down to row 7

Use row 1 for headers. Status 1 is an event, and status 0 is censoring.

Optional spreadsheet formulas

Cell	Formula	Purpose
F2	=COUNTIF($B$2:$B$9,">="&E2)	At risk before time
G2	=COUNTIFS($B$2:$B$9,E2,$C$2:$C$9,1)	Events at time
H2	=COUNTIFS($B$2:$B$9,E2,$C$2:$C$9,0)	Censored at time
I2	=1-G2/F2	Survival factor
J2	=PRODUCT($I$2:I2)	Product so far

Fill down to row 7. The >= rule follows the stated tied-time convention and assumes observation from time zero.

Optional spreadsheet results and step-plot coordinates

Time	2	3	4	5	6	8
At risk	8	7	6	4	3	2
Survival	0.8750	0.8750	0.5833	0.5833	0.3889	0.3889

An XY step plot repeats each event time with the old and new survival values. Begin at (0,1) and stop at day 8.

An ordinary line between event points adds diagonal segments that misrepresent the estimate.

The main formulas

$$Y = \min(T, C), \qquad S(t) = P(T > t), \qquad F(t) = 1 - S(t)$$

$$\widehat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$$

$$P(T > b \mid T > a) = \frac{S(b)}{S(a)}, \qquad q_{a,b} = 1 - \frac{S(b)}{S(a)}$$

For b > a and S(a) > 0. Define the event and population before interpreting any expression.

Source notes and method references

Day 3 Detailed Teaching Notes, 25 pages. Slide footers identify the relevant original pages. All teaching datasets and business scenarios are synthetic.

UCLA Office of Advanced Research Computing, Introduction to Survival Analysis in R
https://stats.oarc.ucla.edu/wp-content/uploads/2025/02/survival_r_full.html

lifelines documentation, Estimating univariate models
https://lifelines.readthedocs.io/en/latest/Survival%20analysis%20with%20lifelines.html

The deck uses survival after events at time t, with event-time factors included for ti <= t.

Further reading on comparisons and event rates

lifelines documentation, More examples and recipes
https://lifelines.readthedocs.io/en/latest/Examples.html

NIST/SEMATECH e-Handbook, Failure or hazard rate
https://www.itl.nist.gov/div898/handbook/apr/section1/apr123.htm

These references support the standard methods. The supplied Day 3 notes provide the teaching examples and numerical exercises.